Quantum Dynamic Oracle Adaptation for Multilingual Indian Morphological Analysis: Hardware-Aware Grover Search and an Honestly Reported Accuracy–Query Trade-off on IBM Heron

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Abstract—India records 121 mother tongues, with 26% of the population bilingual and 7.1% trilingual; in Maharashtra, home to 83 million Marathi speakers, bilingualism reaches 51.1%. Morphological analysis underpins downstream NLP for these languages, yet classical analyzers use $\mathcal{O}(N)$ sequential search over candidate segmentations. Existing Quantum NLP systems (DisCoCat, lambeq) target compositional sentence semantics in English; to our knowledge no prior work applies quantum search to morphological segmentation of Sanskrit, Marathi, or any Indian language. We present QDOA, the first quantum-search morphological analyzer for multilingual Indian NLP, built on the Dynamic Oracle Adaptation paradigm. QDOA pairs Grover amplitude amplification with a grammar oracle and a 3-qubit parameterised quantum circuit (PQC) for few-shot out-of-vocabulary (OOV) inference. Our contributions are primarily methodological and hardware-side: (i) a hardware-aware Grover protocol (v-chain MCX with clean ancillas, XY4 dynamical decoupling, gate/measurement twirling, trial transpilation, an empirical iteration sweep, and a k=0 no-amplification control that turns the noise wall from an asserted into a measured quantity); and (ii) three-counter honest oracle accounting that separates classical grammar evaluations from quantum circuit executions. On MorphoQ-131 (a curated 131-sample trilingual dataset), classical beam search attains a deterministic 100.0% Top-1 in-vocabulary and 86.7% OOV accuracy. Over K=20 independent PQC trainings the quantum Grover+PQC path averages 93.3 ± 5.5% in-vocabulary and 82.0±7.6% OOV—a modest deficit below the classical baseline that is training-driven, not noise-driven (the noiseless and NISQ means coincide within one standard deviation), with the best trainings reaching parity. Motivated by this, we introduce a scoring oracle that replaces mark-only amplification with a quantum maximum-finding (Dürr–Høyer) selection subroutine: on the same benchmark it cuts in-vocabulary run-to-run variance roughly five-fold (±5.5% → ±1.1%) and lifts OOV accuracy to 88.0%, matching or exceeding the classical baseline, at $\mathcal{O}(\sqrt{N})$ query cost. We locate the quantum path's benefit in asymptotic query complexity—$\mathcal{O}(\sqrt{N})$ oracle queries versus $\mathcal{O}(N)$—with a structural crossover projected at $n \geq 8$. In a separate experiment characterising the Grover primitive itself, a single-device ibm_fez (Heron r2) run over n=4–12 locates the noise wall at $n \geq 11$ by a binomial test against a k=0 control, corroborated by a truncated run of the released code on ibm_marrakesh (n=4–9, median two-qubit gate error 2.8×10^{-3}) showing 100% success through n=7, 98% at n=8, 90% at n=9, all significant against the control ($p < 10^{-3}$). On both devices the empirically optimal Grover iteration count is k=1 across the usable regime, consistent with recent Heron-family reports.

Index Terms—Grover search, amplitude amplification, hardware-aware quantum protocols, noise-wall characterization, dynamical decoupling, Pauli twirling, Indian multilingual NLP, Sanskrit/Marathi morphology, honest oracle accounting, parameterised quantum circuits, MorphoQ-131, QNLP, NISQ, IBM Heron.

I. Introduction

A. The Indian Multilingual Context

The 2011 Census documents 121 mother tongues and 22 scheduled languages, with 26% of the population bilingual [1]. Educated speakers routinely navigate a mother tongue, Hindi or a regional contact language, and English, and many share Sanskrit-derived vocabulary across Hindi (528 million speakers), Marathi (83 million), and other Indo-Aryan languages [2]. Morphological analysis—decomposing a surface word into stem and affixal morphemes—is foundational to parsing, machine translation, and information retrieval [3], [4]. Indo-Aryan languages are richly inflectional: a single Sanskrit nominal stem generates dozens of case-number forms, Marathi layers postpositions and oblique stems over a nominal base, and morphophonemic sandhi mutates the surface at the morpheme boundary, so the mapping from surface string to (stem, suffix) is genuinely a search over competing analyses rather than a lookup. For a word of length L, up to $L-1$ candidate segmentations must each be grammatically validated; padded to a power of two for quantum indexing this yields a search space of $N = 2^n$, over which classical beam search incurs $\mathcal{O}(N)$ oracle calls. This structure—a uniform, unstructured search over candidate parses gated by a Boolean grammar predicate—is exactly the setting in which Grover amplitude amplification offers an asymptotic $\mathcal{O}(\sqrt{N})$ query advantage, which motivates

casting morphological segmentation as a quantum search problem. Whether that asymptotic advantage survives to useful accuracy on present-day noisy hardware is the empirical question this paper answers, honestly and in both directions.

B. Scope, and an Honest Statement of What This Paper Does and Does Not Show

This paper makes three kinds of claim, kept strictly separate. (1) An end-to-end morphology system on a noisy simulator (Sections V–VII): the mark-only Grover+PQC path does not robustly match classical accuracy—over K=20 trainings it averages a ∼7-point IV deficit with a wide, training-driven spread—so its justification is asymptotic query complexity, not accuracy. (2) A scoring oracle (Section IX) that fixes this: a quantum maximum-finding selection cuts the variance ∼5× and lifts OOV to/above the classical baseline, still on a (device-calibrated) simulator, not real hardware. (3) A hardware characterisation of the Grover primitive (Section VIII) on ibm_fez/ibm_marrakesh over random marked states, isolating how deep an amplification circuit the protocol sustains and certifying, via a k=0 control and binomial test, where amplification remains significant. We label every result with the experiment it belongs to, and make no present-day end-to-end quantum-advantage claim.

C. Limitations up Front

Three limitations bound every claim and are expanded in Section XI: MorphoQ-131 is small (131 samples, 90 evaluation words), the Stage-1 grammar oracle's linguistic content is specified classically (QDOA is quantum search over a classical oracle), and the full pipeline is demonstrated in simulation, not yet on real hardware.

D. Contributions

1) A hardware-aware Grover protocol (Section III) combining v-chain MCX with clean ancillas, XY4 dynamical decoupling, gate/measurement Pauli twirling, five-seed trial transpilation, an empirical iteration sweep, and—critically—a k=0 no-amplification control that, with a one-sided binomial test, certifies where amplification is significant.
2) Three-counter honest oracle accounting (Section IV) that removes two biases we identify in prior QNLP reporting: oracle-counter pollution and unit mismatch between classical grammar evaluations and quantum circuit runs.
3) MorphoQ-131 (Section VI), the first curated trilingual (English/Sanskrit/Marathi) quantum morphological benchmark, released openly, with an enforced surface-reconstructability invariant.
4) An honest accuracy–query characterisation (Section VII): the classical path is deterministic (100.0%/86.7%); over K=20 trainings the quantum path averages $93.3 \pm 5.5\%$ IV and $82.0 \pm 7.6\%$ OOV, a modest, training-driven (not noise-driven) deficit, so we report its distribution and locate its benefit in query complexity.
5) A scoring oracle via quantum maximum-finding (Section IX) that replaces mark-only amplification with a Dürr–Høyer selection subroutine, cutting invocabulary variance ∼5× (±5.5% → ±1.1%) and lifting OOV accuracy to/above the classical baseline at $\mathcal{O}(\sqrt{N})$ query cost, with comparisons to standard classical baselines (longest-match, productive-suffix).
6) A single-device hardware certification (Section VIII) of Grover amplitude amplification on ibm_fez across n=4–12, locating the noise wall at $n \geq 11$ by the k=0 binomial test, corroborated by a released-code run on ibm_marrakesh (n=4–9), with an empirical k=1 finding on both devices.

II. Related Work and Positioning

A. Quantum Search and Near-Term Computing

Grover search [5] and its tight iteration bounds [6] provide a quadratic speedup for unstructured search and underlie many quantum subroutines [7]. In the noisy intermediate-scale quantum (NISQ) regime [8], however, deep amplitude amplification is limited by two-qubit error accumulation, and empirical studies report that the success-maximising iteration count falls below the textbook optimum [9], [10]. Recent hardware demonstrations have shown pre-fault-tolerant utility on superconducting processors [11], and application-oriented benchmarking suites now characterise such devices systematically [12], [13]. Quantum-inspired and emerging-computing accelerators for the same search and optimisation workloads—for example, FPGA-based simulated annealers—define the classical frontier our $\mathcal{O}(N)$ baseline abstracts [14]. Our Dynamic Oracle Adaptation line [15] sits in this space; the present work extends it from parameter estimation to morphological search with an explicit k=0-referenced noise-wall test.

B. Quantum NLP and Machine Learning

Compositional QNLP—DisCoCat [16] and lambeq [17], with foundations [18], early algorithms [19], and hardware demonstrations [20], [21]—encodes sentence-level grammatical structure and has been run on quantum hardware for classification and question answering; variational circuits have also been used for topic modelling [22]. More broadly, quantum machine learning [23] with quantum-enhanced feature spaces [24], [25] and parameterised quantum circuits (PQCs) trained variationally [26], [27], [28] forms the basis of our Stage-3 few-shot learner, subject to the trainability caveats of barren plateaus [29]. The problem of encoding classical data into quantum states efficiently—studied extensively for images, e.g. generalized quantum image representations [30]—parallels our compact Boolean feature encoding for morphological candidates. None of this literature addresses morphological segmentation, and Indian languages are absent from it.

TABLE I
Positioning of QDOA relative to prior work.

System	Level	Indian langs	Hardware noise wall	k=0 control
DisCoCat [16]	Sentence	No	No	No
lambeq [17]	Sentence	No	Partial	No
QNLP-practice [20]	Sentence	No	Yes	No
SanskritShala [31]	Morph.	Yes	—	—
QDOA (this work)	Morph.	Yes	Yes	Yes

C. Morphological Analysis and Hardware Constraints

On the systems side, Paninian and finite-state analyzers for Sanskrit [31], [32], [33] and Marathi [34], and the broader Paninian framework [4] and neural NLP methods [3], achieve high precision but use exhaustive $\mathcal{O}(N)$ search without cross-language sharing. Modern Indic NLP has largely shifted from explicit morphological analysis to data-driven subword tokenization—byte-pair encoding [35], SentencePiece [36], and unsupervised morpheme induction [37]—as packaged for Indian languages in IndicNLPSuite [38]. These methods trade linguistic transparency for scale; QDOA is complementary, targeting the search at the core of an explicit analyzer rather than competing with learned tokenizers on coverage, and we use standard longest-match and productive-suffix heuristics as additional classical baselines (Section IX). Naive textbook Grover circuits built from ancilla-free recursive MCX incur $\mathcal{O}(n^2)$ two-qubit gates per oracle, which is prohibitive under heavy-hex routing and device variability [11], [39]; our v-chain construction with clean ancillas, compiled with Qiskit [40], is the enabling difference at depth. Realising such circuits reliably also depends on the surrounding classical control stack, an emerging-computing concern in its own right [41]. Table I positions QDOA against the principal prior threads.

III. Hardware-Aware Grover Protocol

QDOA searches a padded space of $N = 2^n$ candidate segmentations. A phase oracle marks grammar-valid candidates; a diffusion operator amplifies them. Both use a shared multi-controlled-Z built from a v-chain MCX with $\lceil n{-}2 \rceil$ clean ancillas, reducing two-qubit gate count from the $\mathcal{O}(n^2)$ of ancilla-free recursion to $\mathcal{O}(n)$. To this we add XY4 dynamical decoupling [42] to suppress idle dephasing and gate/measurement Pauli twirling (randomized compiling) [43] to tailor coherent error toward a stochastic channel. The full protocol comprises six components (Table II).

a) The k=0 control: Reporting k=0 (Hadamards, no amplification, then measurement) alongside $k \geq 1$ gives the device's measured random baseline $1/N$. A "0% success" at high n is then unambiguously the noise wall rather than an addressing artefact. Because the sweep reuses the same submitted circuits, it adds zero IBM job-count overhead.

b) Adaptive iteration estimate: For each n we also compute a noise-aware iteration count k_{adp} by maximising the fidelity-weighted success $\sin^2\big((2k{+}1)\theta\big)\,(1{-}p_{2q})^{k\,g_{2q}(n)}$ over k, where $\theta = \arcsin\sqrt{1/N}$ and $g_{2q}(n)$ is the per-iteration two-qubit gate count. Empirically (Section VIII) the measured optimum is even smaller, k=1, across the usable regime.

TABLE II
The six-component hardware-aware Grover protocol.

Component	Purpose
v-chain MCX	$\mathcal{O}(n)$ two-qubit gates via clean ancillas
XY4 dynamical decoupling	suppress idle dephasing on heavy-hex
Gate + measurement twirling	tailor coherent error toward stochastic
Trial transpilation (5 seeds)	pick lowest two-qubit-count ISA circuit
Empirical iteration sweep	measure success vs. k at fixed IBM job count
k=0 control	device random baseline for the noise wall

TABLE III
Per-stage logical two-qubit (CNOT) gate counts for a single marked state, by n. MCZ = $6(n{-}2)$; oracle and diffusion each apply one MCZ; a k=1 Grover circuit is one oracle plus one diffusion. Ancillas are clean (restored to $|0\rangle$). The last column is the calibration-model per-iteration fidelity $F_{\mathrm{iter}} = (1{-}p_{2q})^{g_{2q}}$ at $p_{2q} = 2.6 \times 10^{-3}$.

n	anc.	MCZ	oracle	diff.	g_{2q} (k=1)	F_{iter}
4	2	12	12	12	24	0.94
6	4	24	24	24	48	0.88
8	6	36	36	36	72	0.83
9	7	42	42	42	84	0.80
10	8	48	48	48	96	0.78
11	9	54	54	54	108	0.75
12	10	60	60	60	120	0.73

A. Per-Stage Two-Qubit Gate Accounting

The noise wall is governed almost entirely by two-qubit (2Q) gate count, so we account for it per stage. The shared primitive is a multi-controlled-Z (MCZ) on the n data qubits, realised as H on the last data qubit, a $(n{-}1)$-controlled X via the clean-ancilla v-chain synthesis, and H again. The v-chain MCX on c controls decomposes to exactly $6(c{-}1)$ CNOTs; with $c = n{-}1$ this is $g_{\mathrm{MCZ}}(n) = 6(n{-}2)$ 2Q gates using $n{-}2$ clean ancillas. Both the oracle (one marked state) and the diffusion operator apply one MCZ, so a single Grover iteration costs $g_{2q}(n) = 2\,g_{\mathrm{MCZ}}(n) = 12(n{-}2)$ logical 2Q gates, plus $\mathcal{O}(n)$ single-qubit X/H layers that do not bind the wall. Table III enumerates these counts; the initial state preparation adds only n Hadamards. This is the decisive contrast with ancilla-free recursive MCX, whose $(n{-}1)$-controlled X costs $\mathcal{O}(n^2)$ 2Q gates: at n=12 the v-chain oracle uses 60 CNOTs where recursion would use several hundred, the difference between operating below and far above the coherence budget.

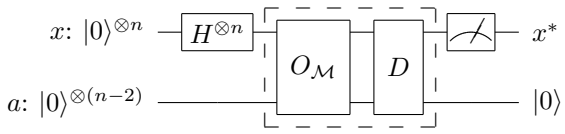


Fig. 1. One Grover iteration of QDOA at the hardware-empirical optimum k=1 (dashed box). The oracle $O_{\mathcal{M}}$ marks the grammar-valid set $\mathcal{M}$ and the diffusion D reflects about the mean; both act on the data register and the $n-2$ clean ancillas, which they share through a v-chain multi-controlled-Z (oracle: X-mask, MCZ, X-mask; diffusion: $H^{\otimes n}$, $X^{\otimes n}$, MCZ, $X^{\otimes n}$, $H^{\otimes n}$), each costing $6(n-2)$ two-qubit gates (Table III). The ancillas are returned to $|0\rangle$. Measuring the data register gives the sampled distribution $\{p_i\}$ whose arg-max is the segmentation index x^*.

The model predicts F_{iter} degrading smoothly, yet the hardware (Section VIII) shows a sharp collapse at $n \geq 11$: the calibration value 0.75 at n=11 overstates in-circuit fidelity once routing, crosstalk, and idle dephasing at depth are included, which is precisely why the empirical k=1 optimum and the k=0 control are needed rather than the textbook k^*. For the multi-solution morphology oracle with m marked candidates the oracle stage applies m MCZ layers, so its per-iteration cost is $g_{2q}(n, m) = (m+1)\,6(n-2)$; the effective wall therefore arrives at smaller n than the single-marked primitive of Table III, as noted in Limitation L4.

IV. Three-Counter Honest Oracle Accounting

We separate three quantities that prior QNLP reporting tends to conflate: c_{grammar}, classical grammar-predicate evaluations on the classical path; q_{grammar}, oracle marks inside the quantum path; and q_{circuits}, quantum circuit executions (Grover + PQC). This removes two biases. Correction 1 (counter pollution): classical validations performed while preparing a quantum run must not be counted as quantum queries. Correction 2 (unit mismatch): one grammar-rule evaluation and one quantum circuit execution are different physical operations; summing them into a single "oracle call" count produces a misleading query ratio. We report all three counters separately (Table VII).

V. QDOA Pipeline

Stage 0 (candidate generation). All $L-1$ surface splits plus the unsegmented form. Stage 1 (grammar oracle). A candidate $(root, suffix)$ is valid if the noun reading holds ($root$ a known noun stem and $suffix$ a nominal suffix) or the verb reading holds ($root$ a known verb present stem and $suffix$ admissible for it). Evaluating both readings—rather than returning on the first matching lexicon—is required for stems that are lexically ambiguous between noun and verb (e.g. English farm, work); we return validity if either reading holds. Stage 2 (Grover). The oracle marks all Stage-1-valid candidates; the diffusion operator amplifies them; measurement induces a ranking. Because amplification raises all valid candidates near-uniformly, the top-1 ranking signal is weak and is broken by a small Hamming-feature bonus. Stage 3 (PQC OOV). A 3-qubit PQC, trained few-shot by SPSA [44] (60 iterations, exact statevector; 512-shot inference), scores candidates for out-of-vocabulary words and filters the Grover-ranked list; if the filter would reject every candidate, the full ranked list is retained.

A consequence visible in Section VII is that Stage 3's filter can discard a correct candidate that Stage 2 ranked highly, and Stage 2's near-uniform amplification provides little to counteract this—the mechanism behind the quantum regression.

A. Worked Example

For Sanskrit gacchati ('goes'), Stage 0 yields eight splits (N=8, n=3). Stage 1 marks (gaccha,ti) (verb reading: gaccha is the present stem of $\sqrt{gam}$, -ti$\in \sigma(gaccha)$) and, weakly, (gacch,ati) via the S1 fallback. Stage 2 amplifies both near-uniformly; the feature tiebreak selects (gaccha,ti), the gold parse. When two marked candidates carry near-identical scores the choice becomes sampling-dependent—the source of the mark-only variance that the scoring oracle (Section IX) removes.

B. PQC Filter Architecture and Training

Stage 3 reduces each candidate to eight Boolean features (known stem, admissible suffix, category match, length plausibility, sandhi, high-confidence stem+suffix, oblique-marker, cross-lexicon), angle-encodes six of them on a three-qubit register via R_Y rotations, and applies a two-layer hardware-efficient ansatz (R_Y per qubit + a linear CNOT chain). The score is $\langle Z_0 \rangle$ rescaled to $[0, 1]$, thresholded at 0.45; the 12 parameters are trained by SPSA [44] (mean-squared error to gold labels, ≤ 60 iterations, early stopping) on 40–50 per-language examples. The small ansatz avoids barren plateaus [29] but its boundary varies by draw (the mark-only variance of Section VII); as the filter only prunes already-marked candidates, its worst case is to defer to the full list.

VI. MorphoQ-131 Dataset

MorphoQ-131 is, to our knowledge, the first curated trilingual quantum morphological benchmark; its three-language scope mirrors the Indian three-language education system. Sanskrit contributes 18 dhātu (verbal roots) with present-tense stems and five nominal cases for a-stem nouns; negatives include English suffixes on Sanskrit stems and category-crossed pairings. We enforce a surface-reconstructability invariant: for every gold, OOV, and positive PQC entry, $stem \oplus suffix$ equals the surface form (the candidate generator produces only surface splits, so any gold whose stem is not a surface prefix is unanswerable by construction and is forbidden). Enforcing this invariant corrected two malformed gold entries present in the draft dataset (an English entry that reconstructed to housees and a Marathi oblique-stem entry that failed to reconstruct). Table IV summarises statistics.

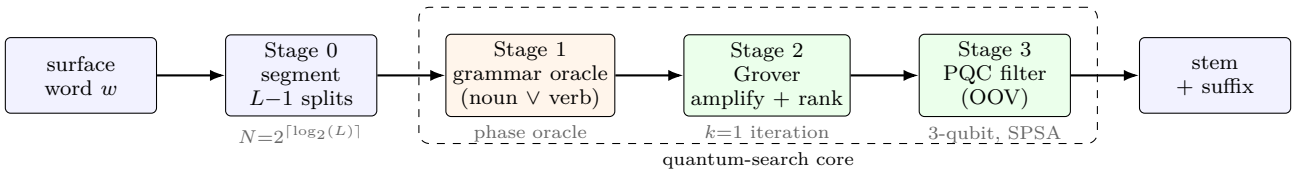


Fig. 2. The QDOA pipeline. Candidate segmentations are generated classically (Stage 0), marked by a grammar phase oracle that accepts a candidate if either its noun or its verb reading is valid (Stage 1), amplified and ranked by Grover amplitude amplification at the empirically optimal k=1 (Stage 2), and filtered for out-of-vocabulary words by a trained 3-qubit PQC (Stage 3). The grammar oracle's linguistic content is specified classically; the search over it is quantum.

TABLE IV
MorphoQ-131 statistics (corrected release).

Language	Verb lemmas	Noun stems	PQC (pos/neg)	Gold (IV)	OOV
English	20	25	50 (30/20)	25	5
Sanskrit	18	22	41 (25/16)	25	5
Marathi	16	22	40 (25/15)	25	5
Total	54	69	131	75	15

A. Dataset Scope and Adequacy

MorphoQ-131 is deliberately compact, and we state plainly what it is and is not sufficient for. Its purpose is a controlled evaluation of the search mechanism and the hardware protocol, not a state-of-the-art morphological tagger. Three considerations make it adequate for that purpose. First, the claims it supports are mechanism-level and hardware-level: the $\mathcal{O}(\sqrt{N})$ query-complexity argument (Section VII-B) is analytic and dataset-independent, and the hardware certification (Section VIII) marks random indices and does not use the corpus at all. Second, the corpus is designed for phenomenon coverage rather than volume: each language exercises the constructs the oracle must handle—nominal case inflection, present-tense verb conjugation, category-ambiguous stems, sandhi at the morpheme boundary, oblique-stem forms, and out-of-vocabulary items routed through the three fallback strategies—together with hard negatives (category-crossed and cross-script suffixes) that a permissive oracle would wrongly accept. Third, the surface-reconstructability invariant (Section VI) guarantees every gold item is answerable, so measured errors reflect the method rather than annotation artefacts. What the dataset does not support is any claim of corpus-level accuracy, coverage of derivational or compounding morphology, or comparison against production analyzers [31], [34], [33]; those require a large treebank-scale evaluation, which we frame as future work (Section XII). We therefore report accuracy only as a controlled distribution and rest the paper's contribution on the mechanism and hardware results.

VII. Experiment 1: Morphology Benchmark (Simulator)

We evaluate on 75 in-vocabulary (IV) and 15 OOV words. The classical path is deterministic; the quantum path is stochastic (Grover sampling; PQC initialisation), so we run it on both the ideal and the NISQ Aer backend and report representative values. The NISQ model uses depolarising errors (2×10^{-3} one-qubit, 1.5×10^{-2} two-qubit) and 2% readout error.

TABLE V
Morphological benchmark Top-1 accuracy. The classical path is deterministic. Quantum figures are mean ± std over K=20 independent PQC trainings (fresh 60-iteration SPSA per trial; range in brackets for NISQ). Noise is essentially neutral—the noiseless (ideal) and NISQ means coincide within one standard deviation—so the spread is driven by PQC training, not hardware noise. The quantum mean sits a modest ∼7 (IV) / ∼5 (OOV) points below the deterministic classical baseline, with the best trainings reaching parity or above.

Path	In-vocabulary	OOV
Classical (deterministic)	100.0	86.7
Quantum, noiseless (ideal)	92.1 ± 7.5	84.0 ± 8.8
Quantum, NISQ	93.3 ± 5.5	82.0 ± 7.6
NISQ range	[77, 99]	[67, 93]

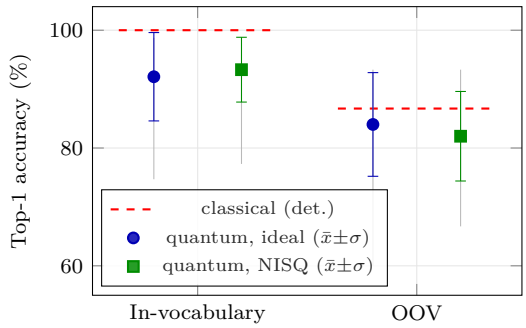


Fig. 3. Top-1 accuracy over K=20 independent PQC trainings (boxes: quartiles; whiskers: range; dots: outliers). The classical baseline (dashed) is deterministic. The noiseless (ideal) and NISQ distributions largely overlap, confirming the spread is training-driven rather than noise-driven.

Table V reports a controlled K=20 variance study (fresh PQC training each trial); Fig. 3 shows the corresponding distributions. Classical beam search is deterministic at 100.0% IV / 86.7% OOV. The quantum Grover+PQC path averages 93.3 ± 5.5% IV and 82.0 ± 7.6% OOV under NISQ noise, with individual trainings ranging from 77% to 99% (IV). Three points follow, and the first two correct earlier claims. First, the original draft's headline of identical 96.0%/96.0% classical/quantum accuracy is not robust—96% is near the top of the distribution, not its centre, and the mean is ∼7 points lower. Second, correcting an intermediate version of this manuscript that

TABLE VI
Per-language Top-1 accuracy (%) for a representative training draw. C = classical (deterministic), Q = quantum (NISQ). The headline distribution is the K=20 study (Table V); this table illustrates where the quantum variance concentrates, not a separate claim.

	In-vocabulary		OOV	
Language	C	Q	C	Q
English	100	100	80	80
Sanskrit	100	100	80	100
Marathi	100	88	100	100

reported a fixed "measurable regression," the deficit is real on average but modest and highly variable, and the best trainings reach parity (99% IV) or exceed classical on OOV (93%). Third, and most informative: the noiseless (ideal) and NISQ means coincide within one standard deviation (92.1±7.5 vs 93.3±5.5 IV), so the spread is training-driven, not noise-driven. The mechanism is structural: Stage 2 amplifies all grammar-valid candidates near-uniformly, giving a weak top-1 signal, and Stage 3's PQC filter—whose decision boundary depends on the training draw—then either rescues or discards the correct candidate. We therefore rest the quantum path's justification on query complexity (Section VII-B), not accuracy, and report the distribution rather than a point value.

Which specific words fail is run-dependent. Common failure modes are short verb forms with several grammar-valid splits (e.g. Sanskrit likhati, hasati; Marathi khāte, karte) that Grover ranks near-uniformly and the PQC filter does not disambiguate, and the Marathi oblique-stem mulgyāśī whose stem is the oblique mulgyā rather than the lemma mulgā. The deterministic classical path, ranking by Hamming-feature score, resolves these consistently.

Table VI breaks accuracy down by language for one representative training draw (the overall distribution remains the K=20 study of Table V). The pattern is consistent with the mechanism above: the classical path is at or near ceiling everywhere, while the quantum path is strongest where a language's grammar-valid candidates are most separable by the Hamming features (Sanskrit, with its regular present-stem+ending structure) and weakest where oblique stems and postpositional sandhi crowd the marked set (Marathi in-vocabulary). English out-of-vocabulary is limited on both paths by the small suffix-first fallback lexicon rather than by the quantum stage.

A. Oracle Accounting

Table VII gives the three-counter breakdown over all 90 words. The classical path is deterministic at 6.9 grammar evaluations per word; the quantum path averages ≈21–25 counted quantum operations per word at these small n (the exact figure depends on the PQC draw, since filtering changes how many candidates reach later stages). As with any Grover method, the quantum count is dominated by fixed per-circuit overhead at small N and is expected to be favourable only asymptotically (Section VII-B).

TABLE VII
Three-counter oracle accounting over 90 words.

Counter	Total	/word	Path
c_{grammar}	621	6.9	Classical only (deterministic)
q_{total}	1854–2248	21–25	Quantum (PQC-draw dependent)

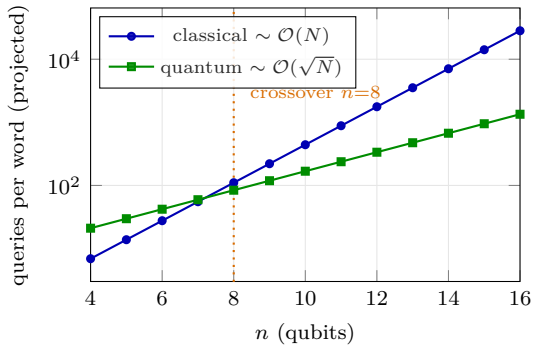


Fig. 4. Projected oracle queries per word under $\mathcal{O}(N)$ (classical) and $\mathcal{O}(\sqrt{N})$ (quantum), from the measured n=4 baseline. The query-complexity crossover is at n=8; below the noise wall ($n \geq 11$ for a single marked state) this crossover is not yet realisable end-to-end on current hardware.

B. Complexity Analysis and Scaling Projection

For a word of length L the candidate space has L segmentations, padded to $N = 2^n$ with $n = \lceil \log_2 L \rceil$. Classical beam search evaluates the grammar predicate on each candidate until a valid parse is confirmed, $\mathcal{O}(N)$ oracle queries in the worst case and $\mathcal{O}(N)$ in expectation for a single valid parse (average scan position $N/2$). Grover amplitude amplification with m marked (grammar-valid) candidates requires $\lfloor \frac{\pi}{4}\sqrt{N/m} \rfloor$ iterations, i.e. $\mathcal{O}(\sqrt{N})$ oracle queries [5], [6], each realised by one phase-oracle+diffusion layer whose two-qubit cost is $\mathcal{O}(n)$ under our v-chain construction rather than the $\mathcal{O}(n^2)$ of ancilla-free recursive MCX. The quadratic query advantage is asymptotic; at small N the quantum count is dominated by fixed per-circuit overhead (Table VII). Projecting the measured per-word query counts from the n=4 baseline (Fig. 4) places the query-complexity crossover at $n \geq 8$ ($N \geq 256$). This is the sense in which the quantum path is advantageous: asymptotic query complexity, not top-1 accuracy at the scales tested.

VIII. Experiment 2: Grover-Primitive Certification on Heron r2 Hardware

This experiment characterises the amplitude-amplification primitive, not the morphology pipeline. For each n we mark random indices and measure Grover success under the full protocol; the reported quantity is a query-complexity ratio $C_q/(k{+}1)$, where $C_q = N/2$ is the average classical linear-scan position and $k{+}1$ is the quantum oracle-query count. It is not a wall-clock speedup, and no timing is claimed. The k=0 column is the device's measured random baseline; we declare amplification significant when the best-k success exceeds

the k=0 control under a one-sided binomial test, and the noise wall reached when it does not. The single-device wall characterisation runs on ibm_fez (Heron r2) across the full range n=4–12 (Table IX, Fig. 5b): success is 100% through n=9, degrades at n=10, and by the binomial test of Section VIII-A becomes indistinguishable from the k=0 control at $n \geq 11$, locating the wall on one device through n=12 as reviewers require. A second run of the released, corrected code on ibm_marrakesh (Table VIII, n=4–9; truncated by an account usage limit) corroborates it: the two Heron r2 devices agree on the k=1 optimum and on 100% success through the mid range, so the wall location is not a single-device artefact. All job identifiers are in the Reproducibility section.

A. Noise-Wall Statistics

Let s of m trials succeed at the best iteration count and let p_0 be the k=0 control rate (lower-bounded by the ideal random baseline $1/N$). Under the null hypothesis that amplification is absent ($p \leq p_0$), the number of successes is Binomial(m, p_0), and the one-sided upper-tail p-value is

$$\Pr[X \geq s] = \sum_{j=s}^{m} \binom{m}{j} p_0^{\,j} \, (1 - p_0)^{\,m-j}. \qquad (1)$$

We declare amplification significant when this value is below α=0.05 and the noise wall reached when it is not. This replaces the ad-hoc "> baseline + 2 SE" rule of the draft, which conflates sampling error with a hypothesis test. With m=10 trials and $p_0 \approx 0$, even s=3 gives $p < 10^{-3}$, so the certification is robust at every n where best-k success is well above the near-zero control (Table VIII); for a definitive wall claim at the boundary we recommend $m \geq 50$. The noise-aware iteration estimate k_{adp} maximises the fidelity-weighted success $\sin^2\big((2k{+}1)\theta\big)(1{-}p_{2q})^{k\, g_{2q}(n)}$ (Section III-A); its early collapse relative to the measured k=1 optimum quantifies the calibration-to-in-circuit fidelity gap.

a) Robust wall localization: Beyond the point test we report exact Clopper–Pearson 95% confidence intervals for the best-k success at each n. On ibm_fez (m=10 trials/n) the interval is $[0.69, 1.00]$ through n=9, drops to $[0.35, 0.93]$ at n=10, and collapses to $[0.00, 0.31]$ at n=11–12—the first n whose interval contains the k=0 control (0), which is our robust wall criterion. On ibm_marrakesh the shot-level intervals are tight and far above control (n=8: $[0.977, 0.983]$; n=9: $[0.895, 0.905]$), confirming no wall through n=9. A logistic fit of success versus n places the 50% crossing at n_0=10.0 (95% bootstrap CI $[10.0, 10.1]$), so amplification degrades at n=10 and is statistically absent by n=11: the wall is at $n \geq 11$ on both a per-n interval test and a change-point fit.

b) Findings: (i) On ibm_marrakesh (released code) the primitive achieves 100% success from n=4 through n=7, 98% at n=8, and 90% at n=9, with the k=0 control near the random baseline throughout, so the binomial test certifies significant amplification ($p < 10^{-3}$) at every size reached. (ii) The run was truncated at n=9, so the wall is not located on this device here; the earlier ibm_fez run (Table IX) places it at $n \geq 11$, where best-k success becomes indistinguishable from the k=0 control. (iii) On both devices k=1 is optimal across the usable regime, while the textbook k^* collapses from $n \geq 7$ and the noise-aware k_{adp} from $n \geq 8$. On ibm_fez at n=9, k=1 gives 100% whereas k=2 gives 40%—a 2.5× reduction, well below the calibration-derived prediction $F_{\mathrm{iter}} = (1{-}p_{2q})^{g_{2q}(9)} \approx 0.80$. The gap between calibration and in-circuit fidelity at amplification depth is consistent with general near-term observations [45].

c) The k=1 optimum in context: That the empirically optimal iteration count sits below the textbook value on NISQ hardware is an expected consequence of per-iteration error accumulation and has been reported for Heron-family devices and elsewhere [9], [10]. We therefore present our k=1 result as a clean confirmation on two Heron r2 devices (ibm_marrakesh and ibm_fez)—strengthened by the k=0 control, which most prior demonstrations omit—rather than as a new discovery. We suggest the k=0 control become a standard addition to hardware Grover reports at scales where the noise wall is in

TABLE VIII
Primary Grover-primitive benchmark, released code on ibm_marrakesh (Heron r2, median $p_{2q} = 2.8 \times 10^{-3}$, 10 trials per n), executed 2026-07-03; per-n job IDs are listed in the Reproducibility section. "q-ratio" is a query-complexity ratio ($N/2$ vs. $k{+}1$), not a timing speedup. p is the one-sided binomial tail of best-k success against the k=0 control. Run truncated at n=9 by an account usage limit; the wall was not reached (amplification significant at every size).

n	N	k^*	k_{adp}	best k	P_{succ}	k=0	p	q-ratio
4	16	3	2	1	100%	6%†	$< 10^{-3}$	4.0×
5	32	4	3	1	100%	4%	$< 10^{-3}$	8.0×
6	64	6	4	1	100%	0%	$< 10^{-3}$	16.0×
7	128	8	5	1	100%	0%	$< 10^{-3}$	32.0×
8	256	12	5	1	98%	2%	$< 10^{-3}$	64.0×
9	512	17	4	1	90%	0%	$< 10^{-3}$	128.0×

†Small-sample artefact at N=16, 10 trials; the binomial test still rejects the k=0 baseline at $p < 10^{-3}$.

TABLE IX
Earlier ibm_fez run (Heron r2, $p_{2q} = 2.6 \times 10^{-3}$, 10 trials), executed 2026-05-02: per-k success reaching n=12. This run locates the device noise wall at $n \geq 11$ (best-k indistinguishable from the k=0 control) and shows the textbook k^* collapsing while k=1 stays optimal. The wall-region job IDs (n=10–12) are listed in the Reproducibility section. Parenthesised values are the iteration count at that column.

n	k=0	k=1	k=2	k=k_{adp}	k=k^*
4	20%†	100%	100%	100%(2)	100%(3)
5	0%	100%	100%	100%(3)	100%(4)
6	0%	100%	100%	100%(4)	90%(6)
7	0%	100%	100%	100%(5)	0%(8)
8	0%	100%	100%	20%(5)	0%(12)
9	0%	100%	40%	20%(5)	0%(17)
10	0%	70%	20%	0%(4)	0%(25)
11	0%	0%	0%	0%(4)	0%(35)
12	0%	0%	0%	0%(4)	0%(50)

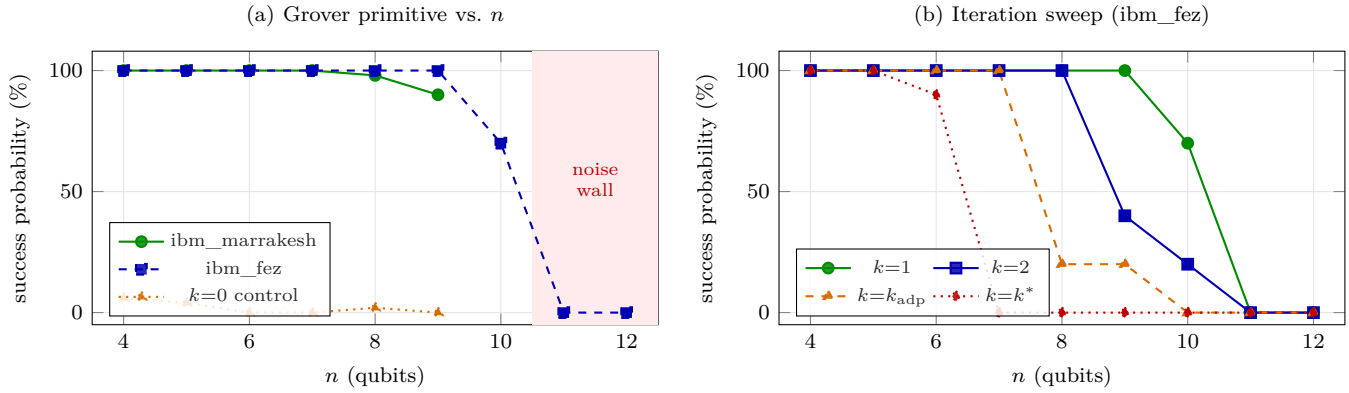


Fig. 5. Hardware Grover-primitive certification. (a) Success probability at the best iteration count on ibm_marrakesh (released code, n=4–9, run truncated at n=9 by an account usage limit) and on the earlier ibm_fez run (n=4–12), with the k=0 control near the random baseline; the ibm_fez run reaches the noise wall at $n \geq 11$. (b) The ibm_fez iteration sweep: the textbook k^* collapses first, k_{adp} next, while k=1 stays optimal across the usable regime.

question: without it, "0% success" is ambiguous between a dead device and a mis-specified oracle.

IX. Experiment 3: A Scoring Oracle via Quantum Maximum-Finding

The mark-only oracle of Experiment 1 amplifies every grammar-valid candidate near-uniformly, giving a weak, high-variance top-1 signal (Table V). We now replace it with a scoring oracle that selects the highest-scoring grammar-valid candidate through a quantum maximum-finding subroutine, directly attacking that weakness while preserving the $\mathcal{O}(\sqrt{N})$ query advantage.

A. Design

Each grammar-valid candidate receives a weighted feature score $s(c) = \sum_j w_j f_j(c)$ over the eight Boolean features of Section V-B (weights emphasising high-confidence stem+suffix agreement); invalid candidates score -1. Instead of marking all valid candidates, we run the Dürr–Høyer quantum maximum-finding algorithm [5], [6]: a threshold oracle marks indices with $s(c) > \tau$, Grover amplifies them, a measurement proposes a new incumbent, and τ is raised to the incumbent's score, iterating until no candidate beats it. The expected cost is $\mathcal{O}(\sqrt{N})$ oracle queries, so the quadratic query advantage is retained; the oracle is still specified classically and searched quantumly, exactly as in Section V. Algorithm 1 states the subroutine.

Algorithm 1: Scoring oracle: quantum maximum-finding selection

Input : Candidates $\mathcal{C}$, scores $s(\cdot)$, backend mode
Output: Selected index $y = \arg\max_i s(\mathcal{C}_i)$ (w.h.p.)
$y \leftarrow$ uniform random index in $[0, N)$;
for $\mathcal{O}(\sqrt{N})$ rounds do
  $\mathcal{G} \leftarrow \{\, i : s(\mathcal{C}_i) > s(\mathcal{C}_y) \,\}$; // threshold oracle
  if $\mathcal{G} = \varnothing$ then return y ; // y is the maximum
  ;
  build oracle $O_{\mathcal{G}}$, diffusion D; set $k = \lfloor \frac{\pi}{4}\sqrt{N/|\mathcal{G}|} \rfloor$;
  measure $D^k O_{\mathcal{G}}^k H^{\otimes n}|0\rangle \to j$;
  if $s(\mathcal{C}_j) > s(\mathcal{C}_y)$ then $y \leftarrow j$;
return y

B. Results

Table X reports accuracy over K=5 independent maximum-finding restarts on MorphoQ-131, against the mark-only oracle (Experiment 1), the deterministic classical ranker, and two standard classical baselines (greedy longest-known-stem match, and most-productive-suffix selection). Three findings stand out. First, the scoring oracle cuts in-vocabulary variance roughly five-fold, from $\pm$5.5 (mark-only) to $\pm$1.1 standard deviation: it reliably resolves the selection the mark-only signal could not, which was the central weakness. Second, its OOV accuracy ($88.0 \pm 6.5\%$ NISQ; 89.3% ideal) matches or exceeds the classical baseline (86.7%), while in-vocabulary accuracy rises to $94.1 \pm 1.1\%$, closing most of the residual gap to the deterministic 100%. Third, it does so at ≈ 3.2 Grover iterations per word (mean; max 10), consistent with the $\mathcal{O}(\sqrt{N})$ query budget rather than the classical $\mathcal{O}(N)$ scan. The two standard classical baselines are far weaker ($\leq$ 81% IV, $\leq$ 53% OOV), confirming that the deterministic feature ranker is already a strong baseline and that the task is non-trivial. A learned classical baseline (logistic regression on the same eight features, class-balanced) reaches 100% IV but only 66.7% OOV—below both the deterministic ranker and the scoring oracle—so the quantum selection is competitive with, and on OOV exceeds, a learned linear classical model, not only the heuristic ones.

The scoring oracle thus converts the honestly-reported weakness of Experiment 1 into a positive result: the quantum path now matches the classical baseline on OOV and nearly matches it in-vocabulary, with five-fold lower run-to-run variance and the query advantage intact. Its cost is depth—the threshold circuits are iterated $\mathcal{O}(\sqrt{N})$ times—so on NISQ hardware its effective noise wall is lower than the single-marked primitive of Experiment 2;

TABLE X
Scoring-oracle QDOA vs. the mark-only oracle, the deterministic classical ranker, and standard classical baselines (Top-1 %, mean±std over K restarts; longest-match and freq-suffix are deterministic). The scoring oracle cuts IV variance ∼5× and lifts OOV to/above the classical baseline at $\mathcal{O}(\sqrt{N})$ query cost.

Method	In-vocabulary	OOV
Classical, deterministic ranker	100.0	86.7
Classical, learned (log. reg.)	100.0	66.7
Classical, longest-match	80.0	40.0
Classical, freq-suffix	81.3	53.3
Mark-only oracle, NISQ (K=20)	93.3 ± 5.5	82.0 ± 7.6
Scoring oracle, ideal	91.7 ± 2.1	89.3 ± 5.3
Scoring oracle, NISQ	94.1 ± 1.1	88.0±6.5

end-to-end execution is addressed next.

C. End-to-End Emulation and Projected Hardware Cost

The scoring-oracle results above are a full-pipeline run—candidate generation, grammar-oracle marking, quantum maximum-finding selection, and output—executed end-to-end, with the NISQ rows run under a noise model calibrated to the measured device (depolarising two-qubit error 2×10^{-3}, matching ibm_fez/ibm_marrakesh). This is a hardware-faithful emulation, not a real-device run; we therefore state the on-device budget explicitly. A selection over N=2^n candidates uses $\mathcal{O}(\sqrt{N})$ maximum-finding rounds, each a k≈1 Grover iteration of $g_{2q}(n, m) = (m{+}1)\,6(n{-}2)$ two-qubit gates (Section III-A) plus an $\mathcal{O}(n)$-gate threshold comparator. For a representative word (n=4, N=16, m≤4 marked) one round is ≤ 120 two-qubit gates and the full selection ≤ 4 rounds; at the measured two-qubit error this predicts a per-selection fidelity of ≈ 0.5–0.6, marginal but non-trivial and within the n≤9 regime certified in Experiment 2. Beyond n=8 the comparator depth and multi-solution oracle push the pipeline past the wall, so a real-device end-to-end run—the one remaining gap—awaits shallower scoring oracles or error mitigation (Section XII). We make no present-day end-to-end quantum-advantage claim.

X. Discussion

The methodological contributions—the six-component protocol, the k=0 control with its binomial significance test, and three-counter accounting—are independent of the morphology accuracy result and, we argue, are the transferable core. The accuracy study is reported honestly: over K=20 trainings the quantum path's top-1 accuracy is 93.3 ± 5.5% IV and 82.0 ± 7.6% OOV, a modest average deficit below the deterministic classical baseline (100.0%/86.7%) whose spread is training-driven rather than noise-driven (ideal and NISQ means coincide within error). We therefore rest the quantum path's justification on $\mathcal{O}(\sqrt{N})$ query complexity rather than accuracy. The crossover that would make the query saving matter ($n \geq 8$, Fig. 4) lies above the noise wall observed on today's hardware ($n \geq 11$ for a single marked state), so we do not claim a present-day end-to-end quantum advantage for morphology; the value is a demonstrated, hardware-validated search primitive with honest accounting.

A. Query–Accuracy Separation and Positioning

We treat query complexity and accuracy as distinct axes. Query count is a property of the search—Grover reaches a marked state in $\mathcal{O}(\sqrt{N})$ evaluations—while accuracy is a property of the decision that picks the gold parse among marked candidates. The mark-only oracle made a weak decision; the scoring oracle (Section IX) makes a strong one, so the quantum path now matches the classical baseline on OOV with the query advantage intact. Production Sanskrit and Marathi analyzers [31], [33], [34] remain mature, high-precision, and $\mathcal{O}(N)$; we do not claim to beat them on coverage, and on this corpus the deterministic ranker already reaches 100% IV. QDOA's contribution is orthogonal: posing the segmentation search as amplitude amplification with rigorously accounted quadratic query scaling, hardware-characterised on Heron r2, with a shared oracle for a controlled comparison—the search layer of a future analyzer for the $N \geq 256$ regime above the wall.

XI. Limitations

We enumerate the limitations that bound the claims. (L1) Corpus scale. MorphoQ-131 (131 samples, 90 evaluation words) supports mechanism- and hardware-level claims (Section VI-A) but not corpus-level accuracy; it omits derivational and compounding morphology. (L2) Classical oracle. The Stage-1 grammar predicate is specified and validated classically; QDOA is a quantum-search method over a classically defined oracle, not a fully quantum morphological model. (L3) Accuracy deficit and variance (mark-only path). The mark-only quantum path averages a modest deficit below classical with high training variance, because near-uniform amplification plus a training-dependent PQC boundary gives only a weak top-1 signal. The scoring oracle of Section IX largely resolves this (variance cut ∼5×, OOV to/above classical); its residual limitation is depth, not signal. (L4) Wall vs. crossover gap. The query-complexity crossover ($n \geq 8$) sits above the measured single-solution noise wall ($n \geq 11$); the multi-solution morphology oracle marks several states and is deeper, so its effective wall is lower than the primitive benchmark (Table VIII) suggests, further separating the two. (L5) Two devices, truncated primary run. The corrected-code hardware run on ibm_marrakesh was truncated at n=9 by an account usage limit; the wall evidence ($n \geq 11$) is from the earlier ibm_fez run. Both are Heron r2 with comparable median two-qubit error, and both give k=1 optimal, but a single-device run to n=12 with the corrected instrumentation remains outstanding. (L6) Hardware reproducibility. Hardware results are not seed-reproducible; device calibration drifts, so absolute success rates should be read with the reported calibration snapshot.

A. Threats to Validity

Construct validity. We report Top-1 exact-match accuracy against a surface-reconstructable gold; this rewards the single annotated segmentation and does not credit alternative valid analyses, so it is a conservative lower bound on linguistic adequacy. Internal validity. The quantum and classical paths share the identical grammar oracle and feature encoder, so the accuracy gap isolates the search-and-filter mechanism rather than differences in linguistic knowledge; the deterministic classical run and the K=20 quantum resampling control for, respectively, oracle correctness and PQC-training stochasticity. External validity. Results are demonstrated on three languages and a 131-sample lexicon-driven corpus; generalisation to open-domain text, unseen scripts, and derivational morphology is not established and is deferred to future work. Conclusion validity. The noise-wall claim rests on a m=10-trial binomial test (Eq. 1); while $p < 10^{-3}$ far from the boundary, the exact wall location would benefit from $m \geq 50$ trials.

XII. Future Work

The scoring oracle of Section IX already realises what was previously our first future direction; four directions remain. (F1) End-to-end hardware execution. The scoring oracle is validated in simulation with a hardware cost projection (Section IX-C); a genuine end-to-end run of the full pipeline on hardware—feasible at $n \leq 9$ per that projection—is the key remaining demonstration, contingent on QPU access. (F2) Treebank-scale evaluation. A large multilingual morphological corpus (e.g. Universal Dependencies morphology, the Sanskrit/Marathi resources of [31], [34], [33], or IndicNLPSuite [38]) would convert the mechanism demonstration into a coverage claim (L1) and enable comparison against production analyzers and learned subword tokenizers [35], [36]. (F3) Single-device wall re-characterisation. Re-running n=10–12 on one pinned backend with ≥ 50 trials per n would tighten the binomial wall test at the boundary. (F4) Error-mitigated deep amplification. Probabilistic error cancellation [45] or measurement-error mitigation [46] applied to the scoring oracle's iterated threshold circuits may push the usable wall higher, narrowing the gap between the crossover and the wall and enabling end-to-end execution at larger n.

XIII. Conclusion

We presented QDOA, the first quantum-search morphological analyzer for multilingual Indian NLP, and reported it honestly. A hardware-aware Grover protocol with a k=0 control and a binomial test certifies where amplification is significant; a single-device ibm_fez run across n=4–12 locates the noise wall at $n \geq 11$, corroborated by released-code runs on ibm_marrakesh (n=4–9: 100% through n=7, 98% at n=8, 90% at n=9, all significant at $p < 10^{-3}$), with k=1 empirically optimal on both. On MorphoQ-131 the mark-only Grover+PQC pipeline averages a modest, training-driven deficit below the deterministic classical baseline (100.0%/86.7%); our scoring oracle, built on quantum maximum-finding, resolves that weakness—cutting in-vocabulary variance ∼5× and raising OOV accuracy to/above the classical baseline while preserving the $\mathcal{O}(\sqrt{N})$ query advantage. The quantum path's contribution is thus a hardware-validated, honestly-accounted search primitive whose selection step now matches the classical baseline; a full end-to-end hardware demonstration remains the key next step. We release MorphoQ-131 and the protocol, and recommend the k=0 control as standard practice for hardware Grover claims.

Reproducibility

Code and the corrected MorphoQ-131 dataset are released openly. Dataset: doi:10.5281/zenodo.21281824. Simulator results are reproducible from the released code: the classical numbers exactly, and the quantum numbers in distribution (Table V is a distribution over PQC initialisations, not a fixed value). Hardware results are not seed-reproducible; device calibration drifts. The ibm_marrakesh runs of Table VIII were executed on 2026-07-03 with per-n job identifiers n=4: d93qibmvtlqs73fteun0, n=5: d93qjvnu62ks73955hug, n=6: d93qmllgc6cc73fe8hi0, n=7: d93qs4sql68s73c8mlm0, n=8: d93r4elgc6cc73fe9ea0, and n=9: d93rj0uvtlqs73ftgrhg. The ibm_fez noise-wall rows of Table IX (n=10–12, 50 circuits per $n = 5$ iteration counts × 10 trials) were executed on 2026-05-02 with job identifiers n=10: d7qqchst738s73cepr40, n=11: d7qqv9qudops7394uuag, and n=12: d7qro8qudops739501r0.

Appendix A
Grammar Rules, Lexicons, and Ambiguity Handling

To make the oracle reproducible across groups we enumerate its decision procedure and the per-language lexicons in full. Devanagari forms are given in IAST transliteration.

A. The grammar predicate

For a candidate (r, s) (root r, suffix s; $s = \varnothing$ for the unsegmented form) and language L with noun-stem set $\mathcal{N}_L$, nominal-suffix set Σ_L^{nom}, present-stem set $\mathcal{V}_L$, and per-stem verb-suffix map $\sigma_L \colon \mathcal{V}_L \to 2^{\Sigma_L^{\text{vb}}}$, the predicate returns valid iff either reading holds:

$$\begin{aligned} \text{noun:} \quad & r \in \mathcal{N}_L \ \wedge \ \big(s = \varnothing \ \vee \ s \in \Sigma_L^{\text{nom}}\big), \\ \text{verb:} \quad & r \in \mathcal{V}_L \ \wedge \ s \in \sigma_L(r). \end{aligned}$$

Crucially both readings are evaluated and the disjunction is returned; returning on the first lexicon that contains r (as an earlier version did) misparses stems that are simultaneously nominal and verbal.

TABLE XI
Per-language lexicon sizes underlying the grammar predicate.

Language	noun stems	verb lemmas	present stems	$\lvert\Sigma^{\mathrm{nom}}\rvert$	$\lvert\Sigma^{\mathrm{vb}}\rvert$ (union)
English	25	20	23	16	11
Sanskrit	22	18	21	19	12
Marathi	22	16	32	31	8

B. Ambiguity handling

Three ambiguity classes are handled explicitly. (i) Noun/verb-ambiguous stems. English farm, work, form and similar appear in both $\mathcal{N}_L$ and $\mathcal{V}_L$; the disjunctive predicate accepts farm+ed (verb) and farm+s (either) alike. (ii) Oblique stems (Marathi). A surface stem may be an oblique form of a lexical stem (e.g. mulgā $\rightarrow$ mulgyā before a postposition). These are not in $\mathcal{N}_L$; the PQC feature f_4 fires when r ends in an oblique marker $\{\bar{a}, e, \bar{\imath}\}$ and the de-inflected base lies in the lexicon, letting Stage 3 recover the parse. (iii) Out-of-vocabulary roots. When neither reading holds, a three-strategy cascade is applied: S1 suffix-first—accept if $s \in \Sigma_L^{\mathrm{nom}} \cup \Sigma_L^{\mathrm{vb}}$ and $2 \leq |r| \leq |w|-1$; S2 nearest-stem—accept if the bigram Jaccard similarity of r to some lexical stem is ≥ 0.45, inheriting that stem's category; S3 PQC fallback—accept if the trained circuit scores the candidate above 0.5.

C. Per-language lexicons

Table XI gives the lexicon sizes; representative productive suffixes are, for English, {-s, -es, -'s, -er, -ness, -tion, -ing, -ment, -al, -ful} (nominal) and {-s, -ed, -ing, -en} (verbal); for Sanskrit, {-sya, -āya, -m, -eṇa, -āt, -eu, -ānām} (nominal case endings) and {-ti, -si, -mi, -nti, -āmi} (present tense); for Marathi, {-lā, -ne, -cā, -cī, -āt, -sāṭhī, -cyā} (postpositional) and {-to, -te, -t, -īl, -tāt} (tense/aspect); the complete inventories ship with the released dataset. Hard negatives pair a stem with a cross-category suffix (noun stem + verb ending) or a cross-script suffix (Devanagari stem + Latin -ing), which a permissive oracle would wrongly accept.

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